

File types & links

Not every name in a directory is the same kind of thing

Dash, dee, and ell

- In a long listing, the first character is the type
- A dash means a regular file
- The letter d means a directory
- The letter l means a symbolic link, and the listing often shows an arrow to the target
- Soft links are names that point at another path

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

File types & links

Read `ls -l` type letters (`- / d / l`), create soft vs hard links, and see what a *broken* symlink looks like.

Starter example: Study `ls -l` — note `d` dirs, `-` files, `l` symlinks. `design_link.v -> design.v` is good; `gone.txt -> missing.txt` is broken. Try `ln -s design.v alias.v` and `ln design.v hard.v`.

Load starter example

Challenges 0 / 22 cleared

Quiz: type -: In `ls -l`, first character `-` means? Answer: file or regular

Answer

Show hintCheckNextIdle

Quiz: type -Quiz: type dQuiz: type lCount dirsCount symlinksBroken link namereadlink goodreadlink brokenCreate soft linkCreate hard linkHard nlinkQuiz: soft vs hardQuiz: hard inoderm soft linkMake a breakNo hard dirRun ls -lToolchain symlinkRelative soft targetQuiz: ls arrowHard survives rmlQuiz: symlink mode

ls -l (lab directory)

Refresh

- regular file d directory l symbolic link
broken target missing

Lab terminal

Starter tree loaded – inspect `ls -l`, then `ln / readlink`

File types and links lab

Real shell practice

```
wsl - module05-file-types-lab/examples/file_types

# Track A - real Unix
# real shell session (Track A)

$ ls -l
total 4
-rwxrwxrwx 1 yongfu yongfu 2152 Jul 18 16:34 README.md
lrwxrwxrwx 1 yongfu yongfu 12 Jul 18 16:34 broken_link -> no_such_file
lrwxrwxrwx 1 yongfu yongfu 10 Jul 18 16:34 my_symlink -> sample.txt
-rwxrwxrwx 1 yongfu yongfu 132 Jul 18 16:34 sample.txt
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 subdir

$ ls -l my_symlink
lrwxrwxrwx 1 yongfu yongfu 10 Jul 18 16:34 my_symlink -> sample.txt

$ readlink my_symlink
sample.txt

$ file sample.txt
sample.txt: ASCII text

$ file my_symlink
my_symlink: symbolic link to sample.txt

$ readlink broken_link
no_such_file

$ ls -l broken_link
lrwxrwxrwx 1 yongfu yongfu 12 Jul 18 16:34 broken_link -> no_such_file
```

Real shell — file types and links

Real shell practice — try these

```
# ls -l — long listing; first character is the type (-, d, or l)  
ls -l
```

```
# ls -l my_symlink — inspect the working symlink line (arrow to  
target)  
ls -l my_symlink
```

```
# readlink my_symlink — print the path the symlink points to  
readlink my_symlink
```

```
# file sample.txt — classify the regular file  
file sample.txt
```

```
# file my_symlink — classify the symlink (follows or reports the link)  
file my_symlink
```

Pitfalls to watch

- Removing a symlink with `remove` deletes the link name
- Do not confuse a broken link with “nothing there”: `readlink` still prints a path
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, practice long listing, readlink, and spotting a broken link
- When you are ready, take the short quiz, then continue to realpath and resolving paths